Answers for "Intro. to Signals & Comm. 2015

1.a. i) $$F(\omega) = \int_{-\infty}^{\infty} f(t)\, e^{-j\omega t}\, dt$$

$$= \int_{-a}^{a} e^{-j\omega t}\, dt$$

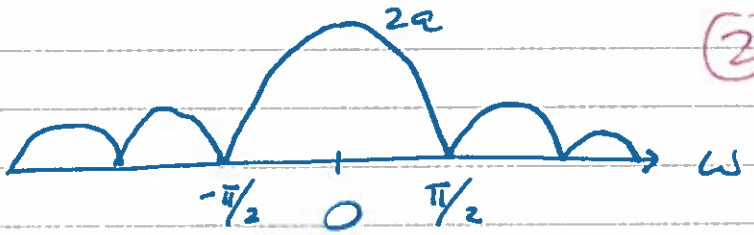
$$= \frac{e^{-j\omega a} - e^{j\omega a}}{-j\omega}$$

$$= \frac{2\sin(\omega a)}{\omega}$$

$$= 2a \cdot \mathrm{sinc}(\omega a)$$

total ↓ (2) [2]

ii) $|F(\omega)|$



(2) [2]

iii) The system is not realizable (1) in practice because of $f(t) \neq 0$ for $t < 0$. That is, it violates the causality property; The system has a response even before the unit impulse is applied to the system. (1)

[2]

1. b. i. $\omega_0 = \frac{2\pi}{T_0}$ (1) [1]

ii. $\int_0^{T_0} e^{jm\omega_0 t} \cdot e^{-jn\omega_0 t}\, dt$

iii) $= \int_0^{T_0} e^{j(m-n)\omega_0 t}\, dt$ (1)

$= \int_0^{2\pi/\omega_0} e^{j(m-n)\omega_0 t}\, dt$

$= \begin{cases} 0 & m \neq n \\ 2\pi/\omega_0 & m = n \end{cases}$ (2) ← [3] (1) [1]

iv. $g(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t}$

$\Rightarrow \int_0^{T_0} g(t) e^{-jm\omega_0 t}\, dt = \int_0^{T_0} \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t} \cdot e^{-jm\omega_0 t}\, dt$ (1)

$\Rightarrow \int_0^{T_0} g(t) e^{-jm\omega_0 t}\, dt = \sum_{n=-\infty}^{\infty} D_n \cdot \int_0^{T_0} e^{j(n-m)\omega_0 t}\, dt$

$\Rightarrow T_0 D_m = \int_0^{T_0} g(t) e^{-jm\omega_0 t}\, dt$

$\Rightarrow D_m = \frac{1}{T_0} \int_0^{T_0} g(t) e^{-jm\omega_0 t}\, dt$ (2)

for $\forall\, m$ [3]

1. b. v. Each signal component $e^{jn\omega_0 t}$ represents a sinusoidal carrier at frequency $n\omega_0$. [2]

vi. The Fourier series coefficient D_n represents how closely the signal $g(t)$ resembles the component $e^{jn\omega_0 t}$. [2]

1 c. i. $\phi_{AM}(t) = A\cos\omega_c t + A\,m(t)\cos\omega_c t$ [2]

ii. The AM essentially translates the signal $m(t)$ at baseband to the frequency band surrounding the carrier frequency $2\omega_c$. [2]

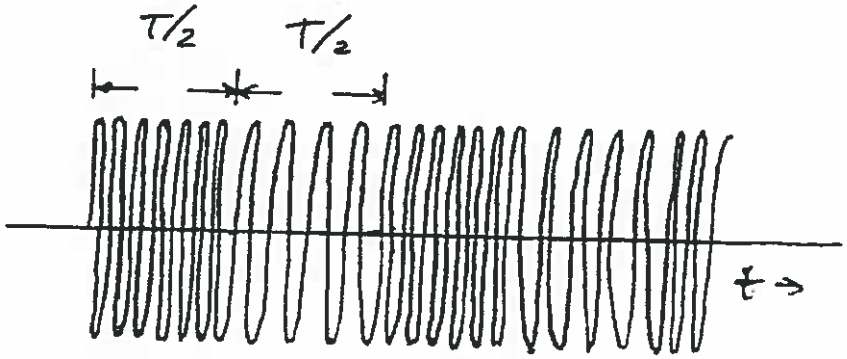
iii. The bandwidth of the AM signal is $2B$ Hz. [2]

iv. $P_c = \frac{A^2}{2}$ [2]

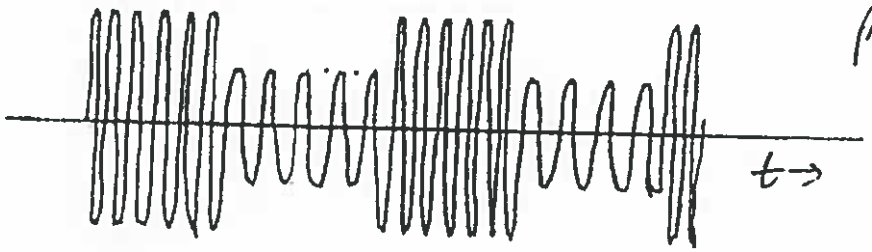
v. $P_s = \frac{E[m^2(t)]}{2} = \lim_{T\to\infty} \int_{-T/2}^{T/2} m^2(t)\cos^2\omega_c t\, dt$ [2]

vi. $\eta = \frac{P_s}{P_s + P_c} = \frac{\frac{E[m^2(t)]}{2}}{\frac{E[m^2(t)]}{2} + \frac{A^2}{2}}$

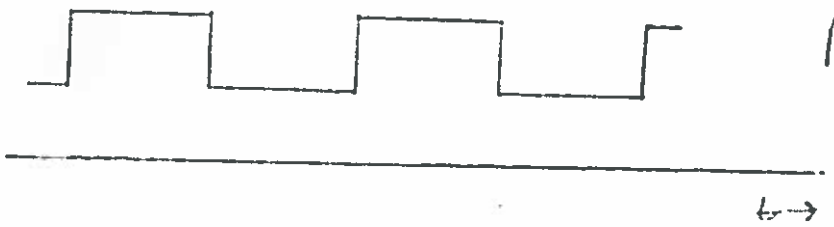
$\eta = \frac{E[m^2(t)]}{E[m^2(t)] + A^2}$ [2]



Point a (3)



Point b (3)



Point c (2)

Point d (2)



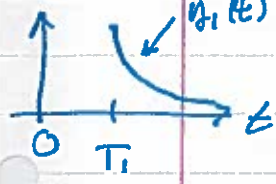
[10]

2. a. i.

$$y_1(t) = \int_{-\infty}^{\infty} x(u)\, h(t-$$

$$y_1(t) = \int_{u=-\infty}^{\infty} h_1(u)\, x(t-u)\, du \quad [3]$$

(2) ii. $$y_1(t) = \int_{u=-\infty}^{\infty} \delta(u - T_1)\, x(t-u)\, du$$



$$y_1(t) = x(t - T_1)$$ (2) due to the property of $\delta(\cdot)$ [4]

ii. The physical effects due to $h_1(t)$ and $h_2(t)$ on the input $x(t)$ are to delay the input by T_1 and T_2 time units, respectively. [2]

iv. $$h(t) = h_1(t) + h_2(t)$$ [2]

v.

$$y(t) = y_1(t) + y_2(t)$$

$$= x(t) * [h_1(t) + h_2(t)]$$ (2)

$$\Rightarrow \quad Y(\omega) = X(\omega) \cdot [H_1(\omega) + H_2(\omega)]$$ (2)

[4]

2. b. i. $\mathcal{F}[f(t-T)] = \int_{t=-\infty}^{\infty} f(t-T)\, e^{-j\omega t}\, dt$ (2)

$\Rightarrow \mathcal{F}[f(t-T)] = e^{-j\omega T} \cdot \int_{t=-\infty}^{\infty} f(t-T) \cdot e^{-j\omega(t-T)}\, dt$

$\Rightarrow \mathcal{F}[f(t-T)] = e^{-j\omega T} F(\omega)$ (2) [4]

ii. The magnitude of the spectrum for $f(t-T)$ does not change, but the phase of the transform does change according to $e^{-j\omega T}$. [2]

iii. $\mathcal{F}[f(at)] = \int_{t=-\infty}^{\infty} f(at)\, e^{-j\omega t}\, dt$ (2)

$= \frac{1}{a} \int_{t=-\infty}^{\infty} f(at)\, e^{-j\left(\frac{\omega}{a}\right)\cdot at}\, d(at)$

$= \frac{1}{a} \int_{t'=-\infty}^{\infty} f(t')\, e^{-j\left(\frac{\omega}{a}\right)t'}\, dt'$

with $t' = at$

$\Rightarrow \mathcal{F}[f(at)] = \frac{1}{a} F\left(\frac{\omega}{a}\right)$ (2)

[4]

2.b. iv. For $a>1$, $f(at)$ is a "compressed" version of $f(t)$ in time. That is, $f(at)$ changes more rapidly when compared with $f(t)$. In that case, $F(\frac{w}{a})$ is "stretched" in w.

That is, one can stretch $F(w)$ in w in order to obtain $F(w/a)$. As a result, more and more energy of $f(at)$ is located in higher frequency regions. (2)

For $a<1$, the opposite is true. That is, $f(at)$ can be obtained by stretching $f(t)$ in time t. Then $F(w/a)$ is "squeezed" in w when compared with $F(w)$. That is, $F(w/a)$ has more and more energy located at lower frequency regions when compared with $F(w)$. (1) [3]

V. When a signal has rapid changes in time, the transform has higher and higher components in high frequency regions. When the signal changes slowly in time, most of its energy will be associated with low frequencies.

[2]

3.a. i. $y(t) = a\phi(t) + b\phi^2(t)$

$\Rightarrow y(t) = aA\cos[2\pi f_{NB}t + \Delta f_{NB}]$

$+ bA^2\cos^2[2\pi f_{NB}t + \Delta f_{NB}]$ (2)

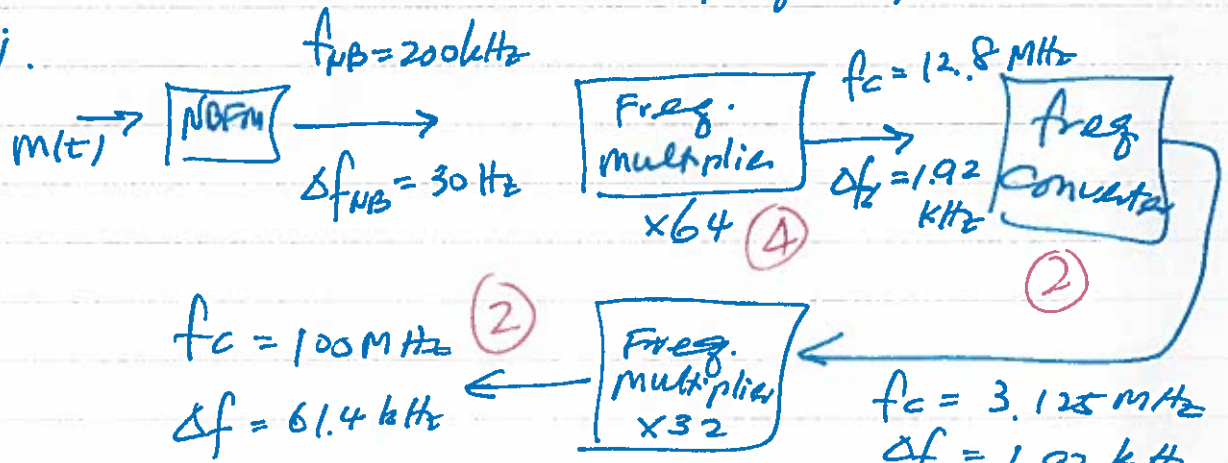
$\Rightarrow y(t) = a\cdot A\cos[2\pi f_{NB}t + \Delta f_{NB}]$

$+ \frac{bA^2}{2} + \frac{bA^2}{2}\cdot\cos[2\cdot 2\pi f_{NB}t + 2\Delta f_{NB}]$ (2)

Double the carrier freq. and frequency deviation.

[4]

ii.



[8]

iii. Oscillator frequency for converter

$9.675\ MHz = 12.8\ MHz - 3.125\ MHz$ (2)

Converter is to multiply a sinusoidal carrier so that the frequency shift can be effected

$\cos(\omega_1 t)\cos(\omega_2 t)$ (4) [6]

$= \frac{1}{2}[\cos(\omega_1 - \omega_2)t + \cos(\omega_1 + \omega_2)t]$

where $\omega_1 = 12.8 \times 2\pi$ rad/sec

$\omega_1 - \omega_2 = 3.125 \times 2\pi$ rad/sec

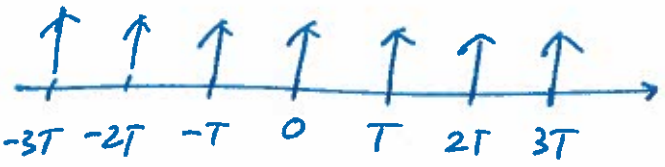
$\Rightarrow$ oscillator freq $\omega_2 = 12.8 - 3.125$

$= 9.675$ MHz

3. b. i. The signaling rate

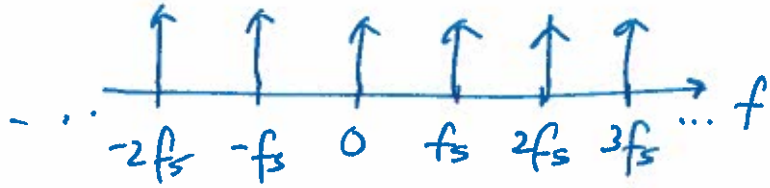
$$f_s \leq 2B$$ [2]

ii. The samples of signal can be interpreted as applying the sampling of pulse train onto a given signal:



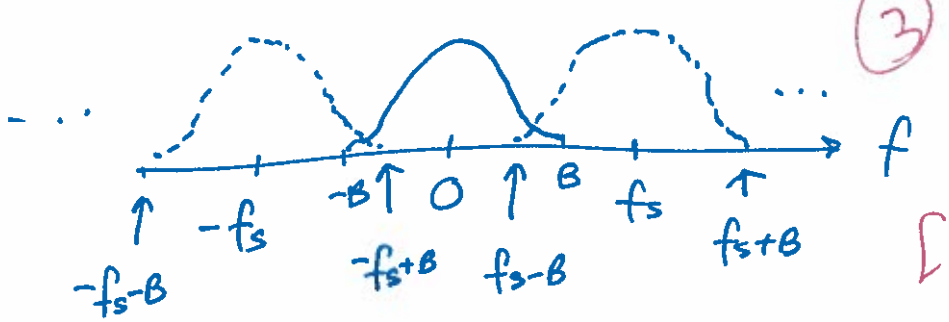
(2)

which has a spectrum of Fourier transform



(2)

As a result, The signal samples of the signal represent a spectrum of



(3)

[7]

From the above diagram,

if $B \geq f_s - B$,

the replica of the baseband spectrum of the original signal do not overlap. That means the original signal has a bandwidth of B Hz or less, which can be supported for transmission by a channel of B Hz. Otherwise, the original signal has components beyond B Hz, which cannot all pass through the channel of B Hz — ~~correct~~

Correct reception is impossible

Overall, the signal rate $f_s \leq 2B$.

iii. $C = 2B \log_2 M$ [3]